AUTOMATED ASSESSMENT OF ACADEMIC PAPERS' GAD RESPONSIVENESS USING NAÏVE BAYES AND BERT

Undergraduate Thesis Project
Submitted to the Faculty of the
Department of Computer Studies of the
Cavite State University-CCAT Campus
Rosario, Cavite

In partial fulfilment
of the requirements for the degree of
Bachelor Science in Computer Science

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October 2025

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An undergraduate thesis project manuscript submitted to the faculty of the Department of Computer Studies, Cavite State University-CCAT Campus, Rosario, Cavite in partial fulfillment for the degree of Bachelor of Science in Computer Science. Contribution No. _____. Prepared under the supervision of Dr. Karlo Jose E. Nabablit and Mr. Vince Lorenz Dela Rea.

INTRODUCTION

Gender mainstreaming has emerged as a central principle in promoting inclusivity achieving fair outcomes within academic and institutional settings. This involves a critical examination of how knowledge production itself can either make or mitigate gender-based inequalities. Consequently, there is a global push for academic outputs to demonstrate genuine gender responsiveness and actively contribute to the goals of Gender and Development (GAD). Institutions are increasingly held accountable to ensure their research output aligns with national and international commitments to gender equality, making the accurate evaluation of academic content a crucial task. Gender mainstreaming ensures that the government pursues equality in development processes and aims for a responsive society where women and men equally contribute (Philippine Commission of Women, n.d.). In the Philippines, the standard bearer for this evaluation is the Harmonized Gender and Development Guidelines (HGDG) (Philippine Commission on Women, 2016). The HGDG serves as the official, comprehensive framework designed to assess the gender responsiveness of government projects, policies, and, critically, research outputs.

Despite the presence of these frameworks, recent studies reveal persistent challenges in evaluating gender inclusivity across research disciplines. Palma (2025) and Sicuan (2024) observed that while higher education institutions have adopted GAD programs, the assessment of research outputs remains inconsistent due to limited awareness, varying evaluator interpretations, and the lack of systematic tools. Similarly, Llorens et al. (2021) and Brück (2023) demonstrated that gender bias continues to shape authorship, citation practices, and publication visibility, showing that inequities persist even in supposedly objective academic environments. These findings highlight that gender bias in academic writing is not only a social issue but also a structural one—embedded in how knowledge is written, cited, and evaluated.

Fortunately, recent advancements in Artificial Intelligence (AI) and Natural Language Processing (NLP) offer a proven solution for objective textual analysis. Researchers such as Mousavian et al. (2025) have shown how machine learning models can identify gendered language patterns and promote fairness in computational systems. Recent research, such as Huy Quoc To et al. (2020), shows that machine learning techniques such as SVM and Naïve Bayes can be used. Meanwhile, Liu (2023) and Onikoyi et al. (2023) explored the use of Naïve Bayes and other classification algorithms in analyzing linguistic patterns, demonstrating their reliability and interpretability in automated text evaluation.

Project Context

Many academic papers demonstrate low or inconsistent Gender and Development (GAD) responsiveness, resulting in research outputs that inadequately address inclusivity, equality, and gender sensitivity. Despite existing guidelines, a significant number of studies fail to integrate GAD principles effectively, leading to insufficiently gender-responsive research from Palma (2025) and Sicuan (2024). Guided by the principles of the Harmonized Gender and Development Guidelines

(HGDG), this study recognizes the need for a systematic, technology-driven approach that can help identify and address gaps in GAD integration, ultimately supporting the production of more comprehensive and responsive academic studies.

Objectives of the Study

The general objective of the study is to develop an automated assessment system that evaluates academic articles based on the Harmonized Gender and Development Guidelines (HGDG) checklist and Naïve Bayes Algorithm.

Specifically, it aimed to:

1. design and implement a model that:
    a. automatically assess academic articles based on the key aspects:
        (1) Gender-Sensitive Language,
        (2) Representation of Women and Men,
        (3) Inclusivity in Language, and
        (4) Authorship and Citations;
    b. classifies textual content using the Naive Bayes algorithm and analyzes contextual meaning using the BERT model;
    c. generates evaluation results and corresponding recommendations; and
    d. provides necessary reports such as assessment summaries.
2. develop the web system prototype that: (1) allows file upload (PyMuPDF and python-docx), preprocesses text (NLTK), performs feature (TF-IDF), performs GAD assessment of academic papers (Naïve Bayes and BERT), and displays results in tabular form (2) uses React.js to provide a responsive and user-friendly web interface (3) use Python (Flask) for backend processing, and (4) use Firebase storing user data, files, and results;
3. test the system in terms of model and functional testing; and
4. evaluate the system using the adapted ISO-IEC 25010 software and data quality evaluation instrument.

Purpose and Description

The primary purpose of this study is to address the inadequate and inconsistent Gender and Development (GAD) responsiveness in academic papers. Many research outputs fail to fully integrate gender-sensitive language, fair representation of women and men, inclusivity, and proper citation practices, resulting in studies that are not fully gender-responsive.

This project aims to provide a systematic and technology-assisted approach to support more comprehensive and inclusive academic writing by identifying gaps in Gender and Development (GAD) responsiveness and offering actionable insights for improvement. In doing so, it benefits authors by guiding them in enhancing the gender sensitivity, inclusivity, and fairness of their manuscripts, helps reviewers and editors maintain consistent and objective evaluation standards across a wide range of submissions, and assists academic institutions in upholding, reinforcing, and further developing their commitment to gender-responsive research. By integrating automated assessment tools with established GAD principles, the system promotes not only improved quality of individual research outputs but also contributes to the goal of fostering a scholarly environment and ultimately, a society that values equity, inclusivity, and the active implementation of gender-responsive perspectives in creating academic papers.

Time and Place of the Study

The study will be carried out from July 2025 to July 2026 at the Cavite State University - CCAT Campus, covering all phases from the preparation of the project outline to the completion of the final manuscript.

Scope and Limitation of the Study

The primary purpose of this study is to develop an automated web-based system that evaluates the Gender and Development (GAD) responsiveness of

academic papers. The system utilizes text classification, specifically the Naïve Bayes Classifier and integrates the Bidirectional Encoder Representations from Transformers (BERT) model to enhance contextual understanding. The main functions of this system include text preprocessing, feature extraction, Naïve Bayes and BERT classification, report generation, and user authentication. The system will also provide features for uploading research papers, viewing responsiveness results, managing user accounts, and accessing assessment summaries for monitoring and evaluation.

For the limitations, this study does not cover the full range of GAD assessment indicators other than the four selected concepts. This includes other dimensions of gender and development evaluation such as participation and decision-making, capacity development, benefits and opportunities for women and men, monitoring and evaluation mechanisms, and sustainability of gender mainstreaming efforts. Additionally, the system does not assess institutional policies, budget allocation for GAD programs, or the broader socio-political context influencing gender responsiveness in research. The focus of this study remains limited to text-based analysis within academic manuscripts, excluding non-textual elements such as images, tables, and reference formatting that may also convey gender bias or imbalance. Moreover, while the system generates automated preliminary assessments and recommendations, final evaluations and decisions remain under the discretion of human evaluators to ensure fairness, accuracy, and comprehensive understanding.

Conceptual Framework

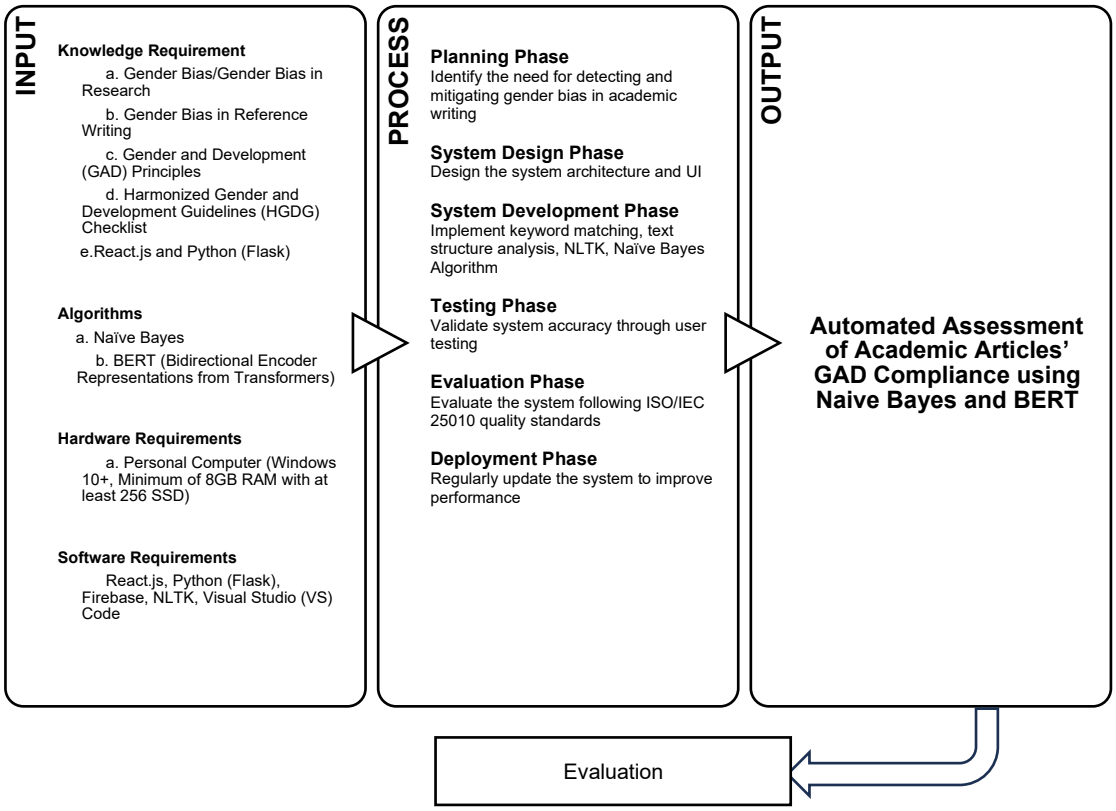


Figure 1. Conceptual framework of the study

The conceptual framework of the study, visually represented by the Input-Process-Output (IPO) diagram in Figure 1, illustrates the flow of activities and components involved in developing the Automated Assessment of Academic Articles' GAD Responsiveness using Naïve Bayes and BERT. The Input consists of key resources such as literature on gender bias, Gender and Development (GAD) principles, and the Harmonized Gender and Development Guidelines (HGDG) Checklist, supported by React.js and Python (Flask) for system development, as well as the Naïve Bayes and BERT models for text classification and contextual understanding. The Process phase follows a systematic approach consisting of planning, system design, development, testing, evaluation, and deployment. During these phases, the system is conceptualized, built, and refined through keyword matching, text structure analysis, and the combined application of Naïve Bayes and

BERT to ensure both linguistic accuracy and contextual fairness in gender-sensitive assessments.

Definition of Terms

The following terms are operationally defined in the study:

Academic Paper refers to a written research output, such as a thesis, journal article, or report, that presents findings and analysis on a specific topic within an academic context.

Gender Bias – refers to the preferential or prejudiced treatment of individuals based on gender, which can influence decision-making, authorship recognition, and citation patterns in academic writing.

Gender-Sensitive Language – refers to a form of communication that avoids bias, stereotypes, or discrimination based on gender.

Gender and Development (GAD) – refers to a framework that promotes equality and inclusivity between women and men by integrating gender perspectives in research, policies, and development programs.

GAD Responsiveness – refers to the extent to which a research paper or project actively responds to and promotes gender sensitivity, equality, and inclusivity principles in alignment with the GAD framework.

Harmonized Gender and Development Guidelines (HGDG) – refers to a set of standardized tools and concepts developed to assess the gender-responsiveness of programs, projects, and research outputs.

Inclusive Language – refers to a language that promotes equality and respect for all individuals by avoiding assumptions or stereotypes related to gender, race, or other social categories.

Machine Learning (ML) – refers to a subset of Artificial Intelligence that uses algorithms to learn patterns from data. It can be applied to detect and correct biased language automatically in academic papers.

Natural Language Processing (NLP) – refers to a field of artificial intelligence focused on enabling computers to understand, interpret, and process human language in text form.

Naïve Bayes Algorithm – refers to a probabilistic machine learning algorithm based on Bayes' Theorem, used for text classification by predicting the likelihood that a document belongs to a specific category.

BERT (Bidirectional Encoder Representations from Transformers) – refers to a deep learning model that analyzes text bidirectionally to understand word context and meaning for tasks like classification and contextual analysis.

REVIEW OF RELATED LITERATURE

This chapter presents the body of existing research and literature relevant to the study, “Automated Assessment of Academic Articles for GAD Responsiveness using Naïve Bayes and BERT.” It explores previous works, frameworks, and technological advancements that form the foundation of the present research.

The growing integration of technology in gender analysis and inclusivity evaluation has led to a surge in studies focused on detecting gender bias, promoting fairness, and automating responsiveness frameworks. Recent work by Mousavian et al. (2025) highlighted how large language models (LLMs) can be leveraged to measure and mitigate gender bias within digital systems, emphasizing transparency and ethical data practices. Similarly, Llorens et al. (2021) found that gender bias in academic writing not only reflects but also reinforces social inequalities, influencing the career progression and well-being of underrepresented groups. These findings show that the problem of bias persists across both human-written and machine-generated texts. Collectively, these studies parallel the need for frameworks such as the Gender and Development (GAD) programs that ensure inclusiveness in institutional and academic outputs. Together, they demonstrate that both computational and institutional approaches are converging toward one goal: the fair and unbiased treatment of gender representation in digital and organizational contexts.

Gender equality and inclusivity remain foundational principles in evaluating development programs and academic work. The Harmonized Gender and Development Guidelines (HGDG), established by the Philippine Commission on Women (PCW, 2020), provide a comprehensive framework for assessing gender responsiveness in planning, implementation, and evaluation processes. Complementing this, the UN Women (2020) rapid assessment tool and its Gender-Responsive Evaluation Handbook (UN Women, 2022) establish systematic ways to

assess equality outcomes across various contexts, highlighting how gender sensitivity can be measured through evidence-based indicators. Similarly, the Commission on Audit (COA, 2020) GAD Audit Guide offers institutional auditing guidelines, while the DBM/PEFA (2024–2025) report focuses on integrating gender-responsive public financial management to ensure inclusivity in budgeting and resource allocation. These frameworks collectively stress accountability and measurable equity in both global and local settings.

Complementing these institutional tools, the Philippine Commission on Women (PCW, 2025) issued the Gender Mainstreaming Evaluation Framework (GMEF), emphasizing self-assessment as a key mechanism for continuous improvement in gender inclusivity programs. However, while UN Women focuses on humanitarian and global programmatic approaches, local instruments like COA's GAD Audit and PCW's GMEF emphasize bureaucratic and institutional compliance—revealing a divergence between international and domestic applications of gender evaluation (UN Women, 2022; PCW, 2025). These converging yet distinct approaches highlight a key challenge: translating policy-level inclusivity frameworks into consistent evaluative and computational methods suitable for automated systems.

In the Philippine higher education context, Palma (2024) emphasized that although many institutions have adopted GAD programs, awareness among faculty and students regarding gender-related laws and policies remains limited. Similarly, Sicuan (2024) reported that while HGDG implementation improves gender awareness and participation in education, significant gaps persist in training and resource allocation. Ray et al. (2022) and Brück et al. (2023) revealed that citation and authorship imbalances remain prevalent, mirroring larger structural inequities that also appear in academic funding and publication. Lee et al. (2023) further showed that gender productivity gaps widened during the COVID-19 pandemic, amplifying these disparities. These studies collectively demonstrate that institutional policies alone are

insufficient, and that automated tools could augment human evaluation by identifying linguistic and structural bias patterns in academic writing.

Negative findings also highlight ongoing gender unresponsiveness in various domains. Trinkenreich et al. (2022) examined challenges faced by women in the software industry and found persistent underrepresentation and lack of inclusive policies that restrict women's participation in technical fields. Similarly, Sagcal and Ramos (2024) investigated GAD program implementation in selected barangays in Nueva Ecija and revealed weak institutionalization and insufficient funding for gender initiatives. These gaps justify the development of automated GAD assessment systems that can complement human oversight, streamline compliance evaluation, and address unresponsiveness in both local and professional contexts.

Recent research underscores that gender-sensitive wording plays a crucial role in shaping inclusivity, perception, and equality in both communication and institutional discourse. Hetjens et al. (2024) demonstrated that job listings written with gender-fair language significantly increased engagement and reduced perceived bias, revealing how linguistic framing can shape audience response and participation. Similarly, Kricheli-Katz and Regev (2021) found that grammatical gender markers unconsciously influence perceptions of competence, resulting in measurable disadvantages for women in professional and academic contexts. These findings converge on the idea that linguistic form carries implicit bias, while diverging in their methodologies, Hetjens et al. focusing on behavioral outcomes, and Kricheli-Katz and Regev exploring linguistic structure. Such studies underscore the necessity of automating the detection and correction of gendered expressions, which manual audits often overlook due to scale and subjectivity.

Beyond individual wording, deeper representational issues persist in the portrayal and visibility of women and men in various fields. Fruehwirth (2024) revealed

that occupational and academic narratives continue to favor men through the use of agentic descriptors such as “leader” or “innovator,” while women are often described with communal or supportive language. Goyanes (2024) further demonstrated that female scholars receive disproportionately fewer citations and lower h-index scores, reinforcing systemic underrepresentation. While these studies differ in focus—one linguistic, one bibliometric—they converge in highlighting that both textual description and citation practice contribute to structural inequity. Addressing these disparities requires systems capable of analyzing both the surface language and deeper contextual structures—a dual capability achievable through models such as Naïve Bayes for lexical detection and BERT for semantic contextualization.

Inclusivity in language extends beyond gender representation alone, encompassing pronoun use, occupational terminology, and exclusionary expressions embedded in everyday writing. Doughman et al. (2021) categorized multiple dimensions of gender bias, including occupational stereotypes, gendered pronouns, and implicit exclusion, and created annotated corpora that enable supervised learning models to detect such patterns. Stańczak and Augenstein (2021) expanded this analysis, showing that NLP models themselves can perpetuate discrimination through binary gender assumptions embedded in training data. Their works converge in exposing algorithmic bias as a byproduct of unbalanced datasets, yet diverge in their solutions—Doughman et al. promote data curation and labeling, while Stańczak and Augenstein call for architecture-level reform. Together, these studies affirm the relevance of fairness-aware models like Naïve Bayes and BERT in advancing automated GAD evaluation tools capable of identifying inclusivity gaps at both linguistic and algorithmic levels.

Even in the sphere of authorship and citation, gender imbalance remains pervasive. Dworkin et al. (2020) identified systematic under-citation of women across scientific literature, a pattern that limits their academic visibility and perpetuates

professional disparities. Wu (2024) corroborated these findings, observing that citation networks consistently favor male scholars, even when research quality is comparable. These converging results reveal how authorship and citation patterns mirror the same biases that appear in language and representation—implying that inclusivity must also extend to credit attribution. Automating the analysis of these patterns through computational tools enables more transparent and data-driven assessments of gender equity within research ecosystems.

To operationalize gender analysis principles, numerous computational and algorithmic innovations have emerged in recent years, bridging linguistic fairness with artificial intelligence. Vanmassenhove, Emmery, and Shterionov (2021) introduced the NeuTral Rewriter, a multilingual Neural Machine Translation (NMT) system built on Transformer architectures, designed to automatically transform gendered English, Spanish, and German sentences into gender-neutral equivalents without losing meaning. The model, fine-tuned with a custom dataset emphasizing gendered text, applied attention mechanisms to guide word substitutions that maintained contextual integrity. Testing involved evaluating BLEU and gender accuracy scores across multiple corpora, and results showed BLEU scores between 27–34 and a 25–40% reduction in gender bias, highlighting the model's effectiveness in generating fluent, gender-neutral text. Similarly, Alhafni et al. (2022) conducted a multilingual shared task on gender rewriting using Transformer-based sequence-to-sequence models, notably ArabicT5, which integrated word-level gender detection for morphologically rich languages. The models generated multiple gender-rewritten alternatives for each sentence to align with gendered person references, and testing on a blind set of 7,318 sentences achieved 98% inter-annotator agreement and 96.2% exact match accuracy. Results indicated strong rewriting performance, with top systems attaining an M2 F0.5 of approximately 88.42, proving the reliability of contextual gender rewriting.

Further developments in modeling semantic fairness stemmed from transformer-based analyses of bias in contextual embeddings. Jentzsch and Turan (2022) employed BERT and its variants to measure latent gender bias by analyzing sentiment shifts between male and female textual variations. Through sentiment analysis tasks across pretrained and fine-tuned BERT models, their testing confirmed consistent polarity differences that traced bias back to pretraining corpora, and results highlighted measurable sentiment asymmetry, proving BERT's tendency to encode gender bias. Complementing this, Dusi et al. (2022) utilized BERT contextual embeddings enhanced by weakly supervised feature selection and dimensionality reduction visualization to map gendered occupational terms in vector space. Their method enabled visual interpretation of gender association clusters, and testing correlated embedding distributions with real-world occupational data, revealing a strong alignment between computational and social gender imbalances—evidence that contextual models mirror real-world disparities.

Efforts toward interpretability in gender bias detection continued through the work of Butt et al. (2021), who fine-tuned BERT and RoBERTa for detecting sexist, exclusionary, and stereotype-driven expressions in social and formal text. Their methodology leveraged attention-based interpretability to link bias predictions to specific tokens. Through evaluation on annotated social media and news corpora, their testing achieved F1 scores between 0.81–0.84, and results showed that attention maps effectively localized biased expressions, enhancing explainability. Expanding this approach, Das et al. (2023) designed explainable transformer pipelines built upon BERT architectures, integrating rationale generation to justify bias classifications. The models produced natural-language explanations for flagged bias, and testing across multiple benchmark datasets yielded 82% accuracy, with human evaluators verifying that explanations were coherent and trustworthy, underscoring the role of transparency in NLP fairness systems.

Beyond textual semantics, Ioannidis et al. (2023) and Kong et al. (2022) explored gender disparity in academia using bibliometric automation. Ioannidis et al. utilized automated author-gender inference algorithms combined with bibliometric network analysis to examine representation among over 100,000 publications. The method inferred author genders and analyzed citation fairness, with testing validated against known author datasets, revealing persistent underrepresentation of women in high-impact networks. Similarly, Kong et al. (2022) applied graph-based algorithms to co-authorship and citation networks, quantifying collaboration patterns and first-mover advantages. Testing on large-scale bibliometric datasets showed systemic disparities in academic visibility, and results confirmed how network structures reinforce gendered recognition gaps—concepts foundational to citation fairness evaluation in the proposed GAD responsiveness system.

At the intersection of linguistic fairness and automated ethics, Sobhani et al. (2024) used context-aware ranking models to assess gender balance in large language model (LLM) outputs. Their method compared text generations across prompts to measure subtle gender imbalance, while testing involved multi-scenario evaluation of ranking fairness. Results demonstrated the model's capacity to identify fine-grained biases using fairness scores rather than classification metrics. Building on this, Kumar et al. (2024) developed the LLM-as-Judge pipeline, which employed advanced LLMs such as GPT-4 and Claude as automated fairness auditors. The approach applied sociolinguistic heuristics to unseen text, evaluating inclusivity with context sensitivity. Testing across various datasets compared model evaluations to human fairness judgments, achieving over 91% alignment accuracy. Results confirmed that large-scale LLMs could replicate human-like fairness reasoning, providing a scalable model for gender responsiveness auditing.

Complementing these modern systems, Bartl et al. (2020) investigated Transformer embeddings (BERT) augmented with counterfactual data augmentation

to reduce gender bias in NLP outputs. Their approach generated gender-swapped text to assess bias resilience, and testing on augmented datasets showed 15–20% reductions in bias metrics. Results validated the method's effectiveness in mitigating bias while preserving performance. Similarly, De Vassimon Manela et al. (2021) employed adversarial debiasing within BERT embeddings to mitigate gender stereotyping during sentence classification tasks. Testing on benchmark datasets revealed a 10–15% decrease in bias scores without loss of accuracy, demonstrating that embedding-level mitigation can yield balanced representations. Meanwhile, Brown et al. (2020) introduced GPT-3, a 175-billion parameter autoregressive transformer, highlighting its unprecedented natural language generation capabilities. Testing through few-shot learning tasks revealed high fluency but also embedded gender and racial biases. Results underscored the duality of power and risk in large-scale models, emphasizing the need for ethical constraints. Extending the scope, Hobbs (2025) employed transformer-based evaluation frameworks to analyze inclusivity in terms of gender and sexual identity. The study benchmarked outputs against annotated datasets representing intersectional identities, with results exposing persistent gaps in LLM fairness across marginalized demographics, urging more intersectionally-aware AI evaluations.

In statistical and probabilistic modeling, Naïve Bayes continues to serve as a robust and interpretable classifier for linguistic bias detection. Angeles et al. (2021) and Tiffani (2020) demonstrated its efficiency in detecting gendered linguistic features and sentiment polarity. Both studies used text preprocessing steps—tokenization and stemming—before training the model on labeled corpora. Testing achieved classification accuracies of 78–82% and ~80%, respectively, validating Naïve Bayes' reliability in gendered language identification. Expanding the comparison, Gupte et al. (2021) evaluated both probabilistic (Naïve Bayes) and neural (LSTM) models on gendered datasets. The study found neural models achieved higher precision (~85%),

but Naïve Bayes maintained better interpretability with ~81% accuracy. Similarly, Liu (2023) introduced hybrid probabilistic-neural classifiers that merged Naïve Bayes and neural networks for multilingual text classification. Testing across languages produced ~79% accuracy for Naïve Bayes and ~83% for hybrid models, reflecting the adaptability of probabilistic methods in cross-lingual bias analysis. Extending this further, Wu (2025) optimized Naïve Bayes with dataset rebalancing, applying it to multilingual corpora to address skewed gender distributions. Results showed a 12% reduction in false negatives and overall accuracy of 81%, enhancing fairness detection. Onikoyi (2023) applied Naïve Bayes for gender inference, predicting authors' gender orientations based on pronoun and linguistic cues. Testing on manually annotated datasets achieved ~80% accuracy, reaffirming Naïve Bayes as an efficient baseline for gender pattern detection.

Several practical applications have also emerged, demonstrating the integration of fairness analytics in real-world systems. As previously mentioned, the NeuTral Rewriter developed by Vanmassenhove et al. (2021) is one of the earliest and most influential systems aimed at automatic gender-neutral rewriting. It combines rule-based grammatical transformations with neural machine translation (NMT) to identify gendered expressions and rewrite them into neutral forms without compromising semantic accuracy. For example, it converts sentences such as "The chairman opened the meeting" to "The chairperson opened the meeting." This hybrid approach effectively integrates symbolic linguistic knowledge with deep learning flexibility, which mirrors the planned design of the current system that merges probabilistic models (Naïve Bayes) and contextual embeddings (BERT). The NeuTral Rewriter's interpretability, achieved through token-level alignment and replacement explanation, also serves as a methodological inspiration for the interpretive interface of the GAD system, ensuring that users can trace why a text was flagged as gender-biased. Sumner (2020) developed the Gender Balance Assessment Tool (GBAT), combining

name-gender inference algorithms and database lookups to compute gender proportions in bibliographies, course syllabi, and citation lists. Testing on real academic datasets verified high accuracy in detecting imbalanced citation practices, and results showed consistent underrepresentation of female authors. Meanwhile, Göritz, Stattkus, and Beinke (2022) introduced a Bias-Detection Dashboard utilizing keyword co-occurrence matrices, sentiment scoring, and heatmap visualization to represent bias patterns in textual corpora. The system, validated against human-annotated corpora, achieved ~90% agreement, offering interpretable visual analytics for gender bias detection. Extending this visualization-driven approach, Kumar et al. (2024) emphasized LLM-based interpretive fairness, proposing an interactive auditing paradigm adaptable for responsive systems. Finally, Dworkin et al. (2020) presented the Gender Citation Balance Index (GCBI), which applied automated name extraction and gender inference to quantify disparities in citation networks. Testing across over 1,000 journals revealed that women authors were cited 20–30% less than their male counterparts, with results confirming systemic citation bias.

In essence, these related systems not only provide theoretical and technical precedents but also serve as modular inspirations for building a comprehensive, automated GAD evaluation platform. The integration of text-level rewriting (from NeuTral Rewriter), gender-balance metrics (from GBAT and GCBI), visualization and interpretability (from Göritz et al.), and scalable fairness assessment (from Kumar et al.) collectively defines a path toward a multi-dimensional system that addresses both the linguistic and structural aspects of gender bias. Such convergence exemplifies how computational intelligence can operationalize inclusivity principles outlined by the GAD responsiveness criteria.

From institutional frameworks to computational systems, research increasingly aligns toward achieving measurable gender fairness and inclusivity in language, representation, and authorship. Different studies approach this goal from varying

directions—some emphasizing policy implementation and social awareness, others focusing on automation and algorithmic precision. Together, these perspectives reveal that genuine gender responsiveness relies on both human insight and technological innovation. Building on this progression, the proposed system integrates machine learning methods such as Naïve Bayes and BERT to evaluate academic papers through the lens of GAD principles, aiming to transform traditional manual assessments into a transparent, data-driven process that promotes equality in academic discourse.

System Technical Background

In writing the technical background, the researcher must discuss the technical aspects of the project being conducted. Their goal is to inform and educate the readers on the technicalities of their project so that they will understand the literature and studies that will be presented in the next section. This includes an in-depth discussion of technical terms involved or used in the project. These terms include jargon or newly coined words in the field of Information Technology.

The Automated Assessment of Academic Papers' GAD Responsiveness with Naïve Bayes and BERT is a web-based system that automatically evaluates the gender responsiveness of academic papers based on four key aspects: (1) Gender-Sensitive Language, (2) Representation of Women and Men, (3) Inclusivity in Language, and (4) Authorship and Citations. It combines natural language processing (NLP) and machine learning techniques to detect non-GAD-responsive content and provide corrective suggestions for improvement. The system draws from research in explainable AI and fairness-aware language models (Mousavian et al., 2025; Sobhani et al., 2024; Kumar et al., 2024), aligning computational tools with the broader goal of promoting inclusivity in academic writing.

Gender and Development Framework Integration

The conceptual foundation of the system draws inspiration from institutional gender evaluation frameworks, such as the Philippine Commission on Women's Harmonized Gender and Development Guidelines (HGDG) and the Gender Mainstreaming Evaluation Framework (GMEF). While the system does not directly apply these frameworks, it takes guidance from their emphasis on inclusivity, equity, and measurable responsiveness in content evaluation. Similarly, global tools such as the UN Women (2020, 2022) Gender-Responsive Evaluation Handbook and the Commission on Audit's (2020) GAD Audit Guide inform the system's goal of standardizing fair, objective assessment practices. By integrating these principles into automated evaluation, the system transforms subjective GAD checking into a more consistent and data-driven process, supporting institutional efforts toward gender fairness in academic writing.

Web-Based Architecture

The system is designed as a web platform developed using React.js, Flask, Firebase, and NLTK. React.js is used to create an interactive user interface where users can upload academic papers and view analysis results. Flask, a Python-based framework, manages backend processes, including NLP preprocessing, model execution, and classification. Firebase serves as the cloud-based database for storing processed data and results, ensuring scalability and accessibility.

Following the design principles of web-based analytical systems proposed by Göritz et al. (2022) and Kumar et al. (2024), the integration of these technologies allows the system to perform real-time gender-responsiveness assessments efficiently while maintaining a user-friendly and secure web environment. This architecture enables seamless interaction between the analytical components and the user interface, ensuring that automated evaluations are both accessible and interpretable to users.

Natural Language Processing and Preprocessing

The preprocessing stage prepares uploaded text for analysis using the Natural Language Toolkit (NLTK). This includes tokenization, stop-word removal, and stemming—steps proven to enhance the accuracy of text-based classification (Angeles et al., 2021; Tiffani, 2020). During this stage, gendered words, pronouns, and exclusionary terms are normalized to allow consistent analysis by the models. Following the approach of Doughman et al. (2021), the preprocessing also ensures that linguistic cues such as role descriptors and gendered associations are properly captured before being evaluated by the Naïve Bayes and BERT classifiers.

Naïve Bayes Classifier (NBC)

Naïve Bayes is a probabilistic text classification model based on Bayes' theorem, which assumes independence among features. Angeles et al. (2021) and Tiffani (2020) applied Naïve Bayes to detect gendered linguistic patterns in academic and social texts. The model calculates the likelihood of a text belonging to a gendered or neutral category based on word frequencies and conditional probabilities. In testing, Naïve Bayes achieved classification accuracies ranging from 78% to 82%, demonstrating efficient identification of gendered expressions. Wu (2025) further improved its performance by tuning the model and rebalancing skewed datasets, reducing false negatives by approximately 12%. Naïve Bayes is well-suited for automated GAD responsiveness systems due to its computational efficiency and interpretability, particularly for lexical-level bias detection and authorship analysis.

Naïve Bayes Algorithm

The Naïve Bayes algorithm functions as one of the core classifiers of the system. Known for its simplicity, interpretability, and effectiveness in text classification (Gupte et al., 2021; Liu, 2023), Naïve Bayes detects surface-level linguistic patterns, such as biased word choices or gender-exclusive terms. In this system, it is applied to assess multiple criteria, particularly Criterion 1, 3, and 4, authorship and citations, as

well as to assist in the identification of gendered or stereotypical phrases. By analyzing word frequency and probability distributions, Naïve Bayes categorizes sentences or phrases as either GAD-compliant or non-GAD-compliant, providing fast and reliable results suitable for real-time feedback.

BERT Model

Complementing Naïve Bayes, the system employs BERT (Bidirectional Encoder Representations from Transformers) for deeper, context-based analysis. As shown in the studies of Dusi et al. (2022) and Butt et al. (2021), BERT effectively captures the relationships between words, enabling detection of subtle biases that occur in sentence structure and meaning. In the system, BERT is specifically used for Criterion 2: Representation of Women and Men, where contextual understanding is essential. For instance, BERT can detect implicit stereotypes or biased portrayals even if gendered terms are not explicitly mentioned. Its contextual embeddings enhance the interpretability and fairness of the system's overall analysis.

Hybrid Model Integration

The system adopts a hybrid architecture combining Naïve Bayes and BERT, inspired by Vanmassenhove et al. (2021) and Alhafni et al. (2022). This dual-model structure leverages the strengths of both approaches—Naïve Bayes for surface-level lexical bias detection and BERT for deeper contextual understanding. The combination improves the system's precision and recall across all four GAD criteria, ensuring that both explicit and implicit forms of gender bias are identified. This integration allows the system to provide both statistical and semantic perspectives in evaluating gender responsiveness.

Neural Machine Translation-Based Gender Rewriting (NeuTral Rewriter)

Vanmassenhove, Emmery, and Shterionov (2021) developed the NeuTral Rewriter, a multilingual Transformer-based NMT system designed to rewrite sentences

into gender-neutral forms. The model was fine-tuned on a custom dataset containing gendered texts in English, Spanish, and German, using attention mechanisms to guide word choices toward neutrality. Evaluations on parallel corpora employed BLEU and gender accuracy scores, achieving BLEU 27–34 and a 25–40% reduction in gender bias. The system preserved grammatical and contextual integrity while neutralizing biased phrasing. In the proposed GAD system, NeuTral Rewriter's approach informs a text-rewriting module that suggests inclusive alternatives to gendered sentences.

Large Language Models (LLMs) / GPT Approaches

Large Language Models such as GPT-3 and GPT-4 are autoregressive transformers capable of generating high-quality text and performing complex natural language tasks. Brown et al. (2020) demonstrated GPT-3's potential and risks, including the propagation of bias in generated outputs. Kumar et al. (2024) introduced the LLM-as-Judge pipeline, where GPT-4 and Claude models acted as fairness auditors by applying sociolinguistic heuristics to unseen texts. Sobhani et al. (2024) employed context-aware ranking models to measure fairness in LLM outputs, while Hobbs (2025) emphasized intersectional representation. These approaches enable adaptive evaluation of gender inclusivity, allowing the system to score text fairness dynamically and provide rationales for detected biases.

Ethical and Fairness Considerations

The system adheres to the ethical AI principles discussed by Mousavian et al. (2025) and Stańczak and Augenstein (2021), emphasizing transparency, fairness, and bias mitigation in data and model design. It employs fairness-aware preprocessing and context-based reweighting to reduce potential algorithmic bias during training. These measures align with the recommendations of Sobhani et al. (2024) and Hobbs (2025), who underscored the necessity of contextual fairness and intersectional sensitivity in NLP systems. Through these strategies, the proposed system not only automates

evaluation but also ensures that its methods remain consistent with ethical standards in AI and gender studies.

System Flow

The system flow begins when the user uploads an academic paper through the React.js front-end, which sends the document to the Flask backend for processing. Using NLTK, the text undergoes preprocessing steps such as tokenization, stop-word removal, and lemmatization to prepare it for analysis. The Naïve Bayes classifier then detects gendered language and citation-related biases, while the BERT model performs contextual analysis to evaluate the fair representation of women and men. The processed outputs—comprising inclusivity scores, representation metrics, and authorship evaluations—are securely stored in Firebase. Finally, the system displays the analyzed results through an interactive visualization table, providing users with automated, data-driven feedback on the GAD responsiveness of their academic papers.

Overall, the Automated Assessment of Academic Papers' GAD Responsiveness system integrates gender fairness concepts with advanced NLP and machine learning techniques. Using Naïve Bayes and BERT, it evaluates key aspects of gender responsiveness within academic papers and provides actionable feedback through a structured, easy-to-understand results table. The system draws inspiration from institutional gender frameworks such as HGDG and GMEF while implementing these principles in a technological context. By combining inclusivity-focused design, efficient NLP preprocessing, and web-based accessibility through React.js, Flask, Firebase, and NLTK, the system offers an automated, scalable, and transparent tool for promoting gender sensitivity in academic writing.

Synthesis

Research on gender responsiveness and inclusivity spans institutional frameworks and computational approaches, reflecting diverse strategies to promote equity. Local guidelines such as the Harmonized Gender and Development Guidelines (HGDG) and the Gender Mainstreaming Evaluation Framework (GMEF) (PCW, 2020; 2025) provide measurable indicators for assessing gender sensitivity, while global instruments like the UN Women (2020, 2022) evaluation handbooks emphasize accountability and evidence-based evaluation. Computational studies, including Vanmassenhove et al. (2021), Göritz et al. (2022), and Kumar et al. (2024), illustrate the potential of NLP and AI in detecting gender bias, linguistic imbalance, and representational disparities. Similarly, Dworkin et al. (2020) and Goyanes (2024) highlight persistent inequities in authorship and citation, while Stańczak and Augenstein (2021) and Mousavian et al. (2025) show algorithmic biases embedded in datasets and models. Although these studies diverge in methodology—ranging from institutional auditing to AI-driven textual analysis—they converge on the shared goal of fostering fairness, inclusivity, and transparency in academic and digital contexts.

Despite these advancements, a significant gap remains in systematically evaluating gender responsiveness in academic writing. Many scholarly articles continue to exhibit gendered language, representational imbalance, and citation inequities, reflecting the inadequacy of current studies and tools to ensure GAD compliance. Addressing this gap, the proposed system integrates Naïve Bayes and BERT to assess linguistic and contextual features aligned with GAD criteria, including gender-sensitive language, inclusivity, representation of women and men, and authorship and citations. This approach seeks to provide an automated, data-driven solution that enhances equity and inclusiveness in academic publications.

METHODOLOGY

This chapter describes the structured approach followed in designing, developing, and testing the proposed system for the automated assessment of academic articles' Gender and Development (GAD) responsiveness. It explains the methodological framework adopted in creating the *Automated Assessment of Academic Articles' GAD Responsiveness using Naive Bayes and BERT*, which utilizes natural language processing and machine learning methods to analyze and classify text. The development process involves data collection, text preprocessing, feature extraction, and model training. The Naïve Bayes algorithm is employed for probabilistic text classification, while BERT (Bidirectional Encoder Representations from Transformers) is utilized for contextual analysis, particularly for evaluating the "Representation of Women and Men" criterion.

Overall, this methodology establishes an organized, data-driven procedure for building an intelligent assessment tool that supports gender responsiveness in academic writing while validating the effectiveness and accuracy of Naive Bayes in automated text evaluation.

Design of Software, Systems, Product and/or Processes

Figure 2 presents the hierarchical input-process-output diagram of the system.

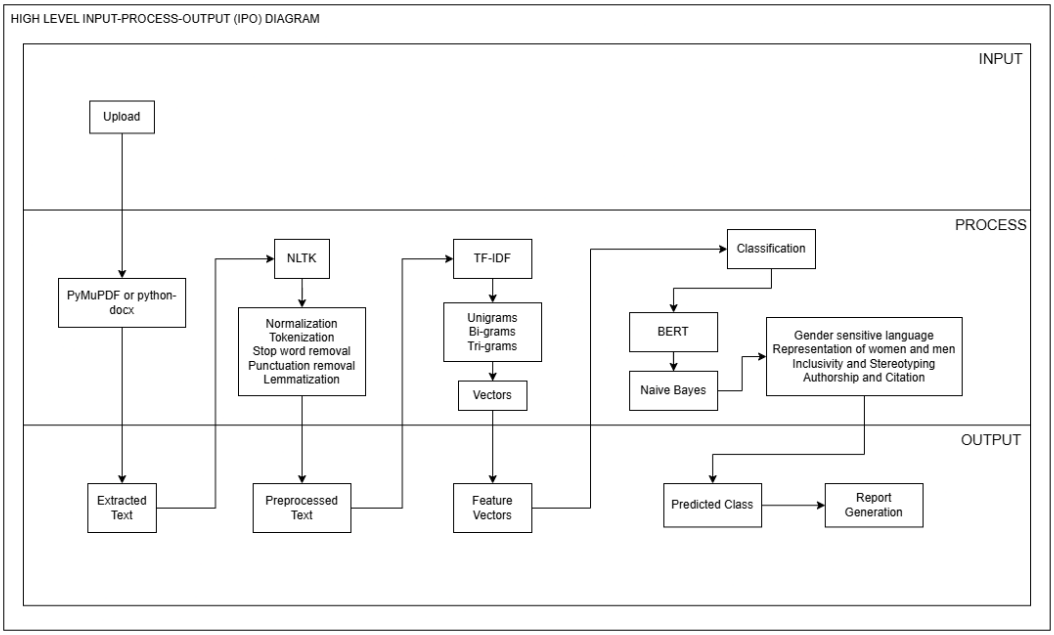


Figure 2. IPO diagram of the system

Figure 3 shows the use-case diagram of the system.

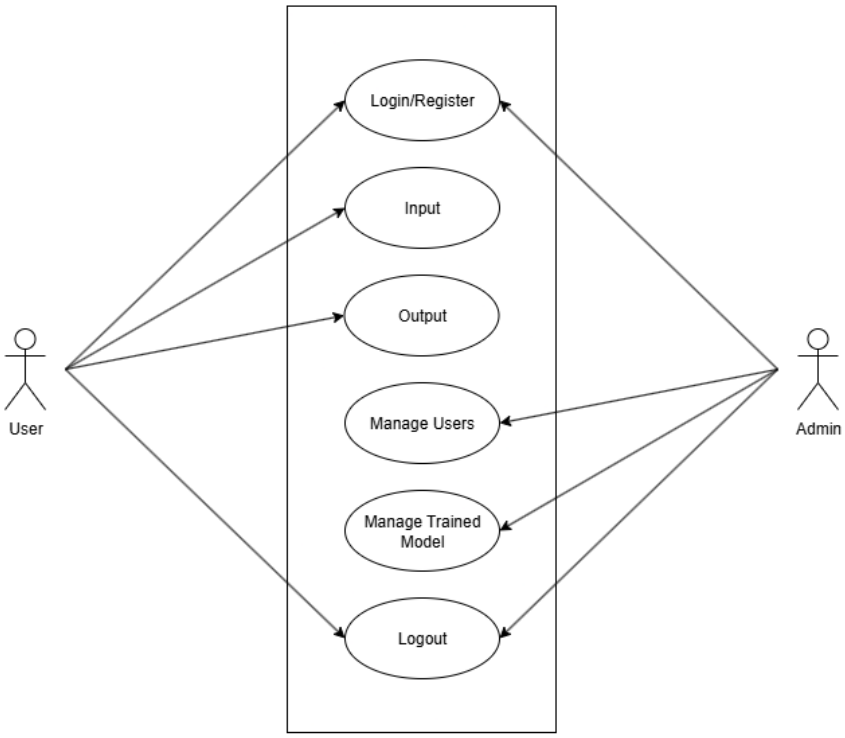


Figure 3. Use-case diagram of the system

The following figures shows the graphical user interface design of the system.

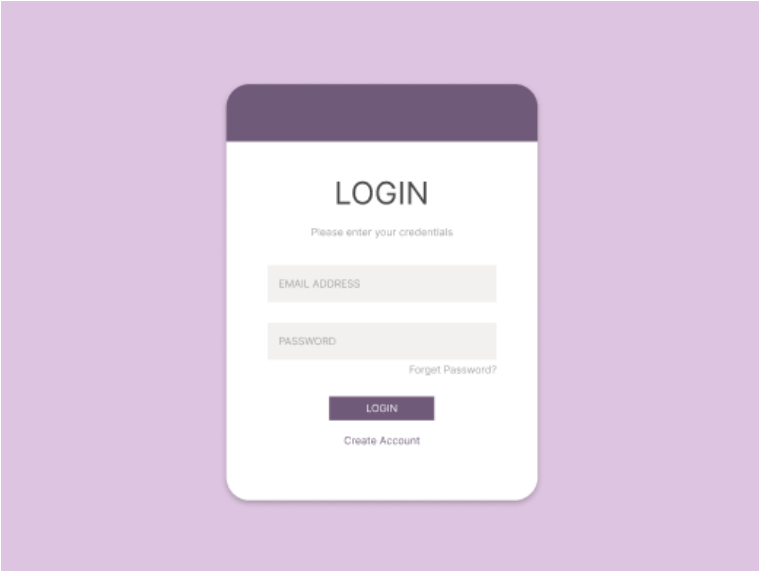


Figure 4.1. Login page

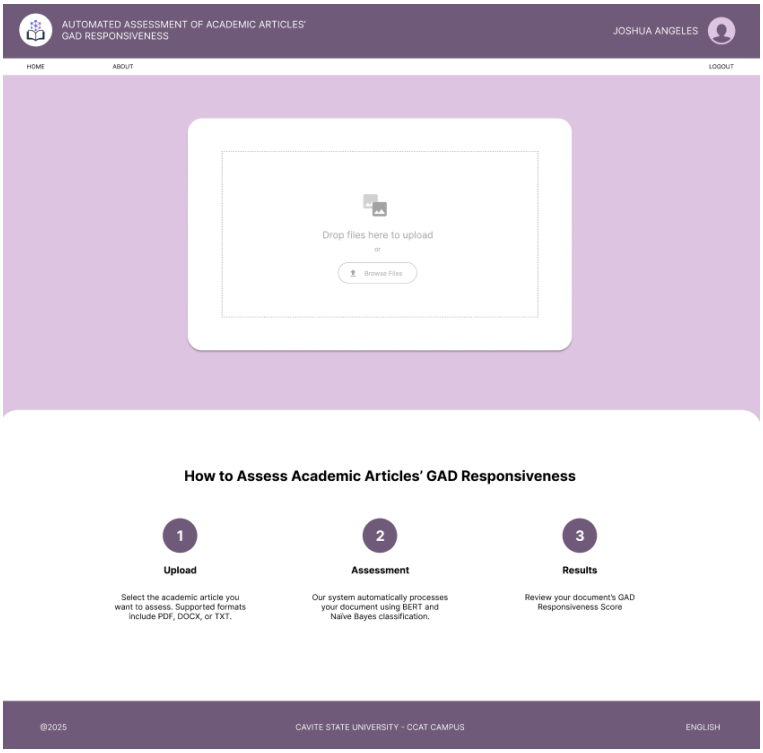


Figure 4.2. Dashboard

Figure 4.3. Output prototype

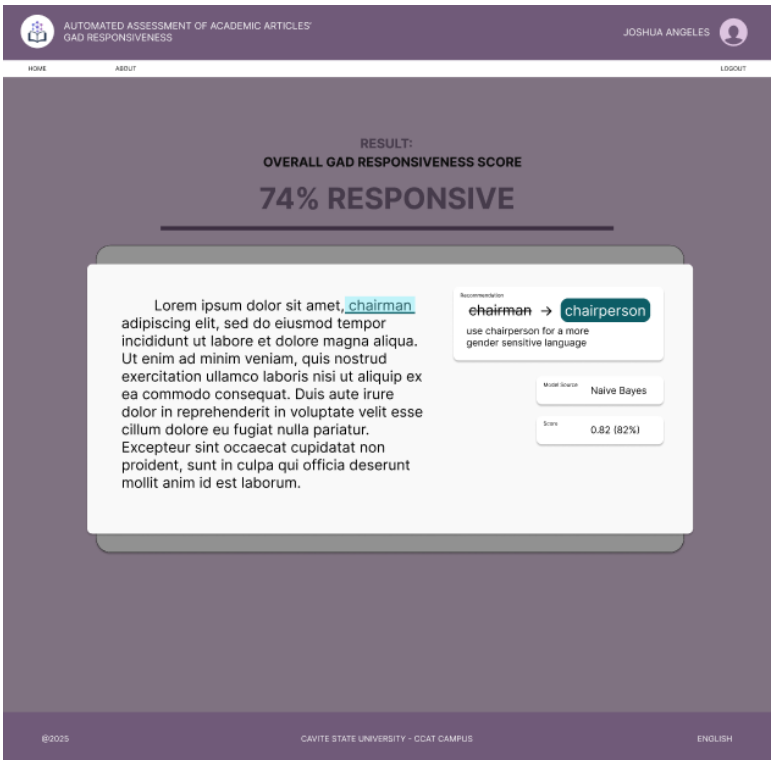


Figure 4.3. Recommendations preview

Requirement Analysis

The Gender and Development (GAD) framework serves as a foundation for promoting gender equality and inclusivity across various sectors, including education and research. Currently, many academic papers remain non-gender-responsive, exhibiting biases in language and representation. Furthermore, the inadequacy of gender-responsive research outputs highlight the need for systems that can guide and evaluate inclusivity in academic writing. The *Automated Assessment of Academic Articles' GAD Responsiveness using Naive Bayes and BERT* addresses these challenges by utilizing machine learning with the help of the HGDG concepts to identify and classify gender-related patterns in text automatically.

Figure 5 presents a fishbone diagram that shows the operational feasibility of the system.

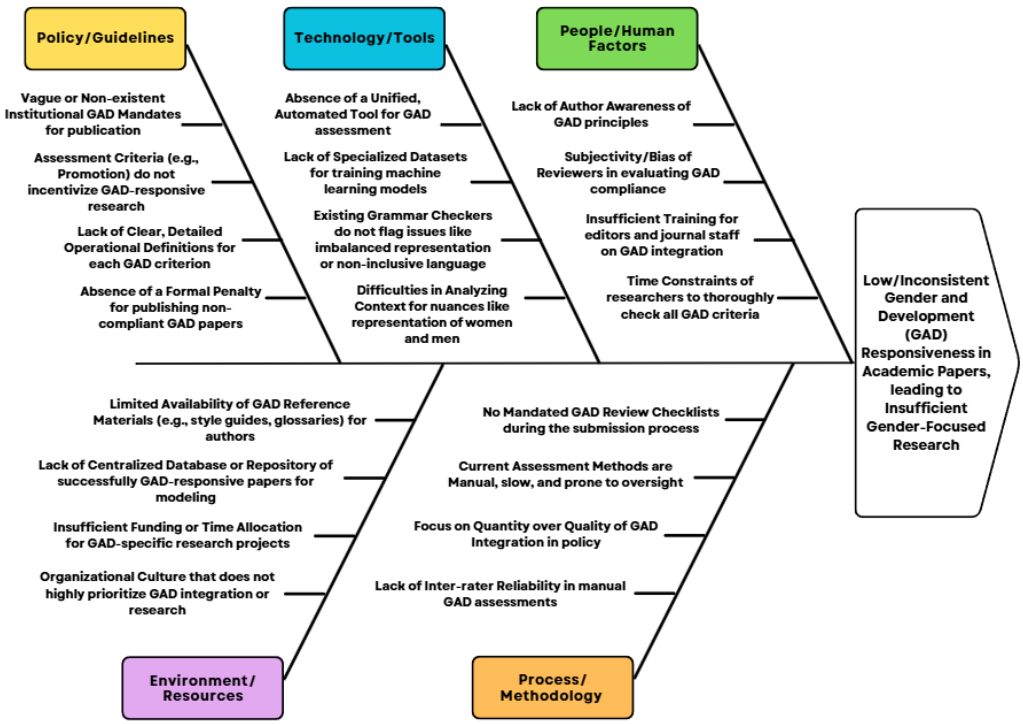


Figure 5. Fishbone diagram of the system

The fishbone diagram illustrates the factors contributing to low or inconsistent Gender and Development (GAD) responsiveness in academic papers, which results in

insufficient gender-focused research. It categorizes causes into five main areas: Policy/Guidelines issues such as vague mandates and weak enforcement; Technology/Tools limitations, including the lack of automated assessment tools and relevant datasets; People/Human Factors, such as limited author awareness, reviewer bias, and insufficient training; Environment/Resources, including scarce GAD reference materials and insufficient funding; and Process/Methodology weaknesses, like the absence of standardized review checklists and reliance on slow, manual assessments.

The proposed web system directly addresses the issues identified in the fishbone analysis. The system integrates automated text processing, feature extraction (TF-IDF), and classification models (Naive Bayes and BERT) to replace manual evaluations with data-driven analysis. The system assists authors and evaluators by offering clear, objective feedback on GAD responsiveness, reducing bias and improving awareness of gender-sensitive writing. It centralizes GAD-related evaluation processes in one accessible platform, minimizing the need for external references and manual cross-checking. Lastly, it streamlines file upload, preprocessing, and result generation into an efficient workflow, ensuring faster, more consistent, and reliable GAD assessments. Overall, the system functions as an intelligent, automated tool that promotes fairness, accuracy, and inclusivity in academic writing.

Table 1 below presents a summary of the key aspects considered in the manual assessment of GAD responsiveness.

KEY ASPECTS	**DESCRIPTION**
Gender Sensitive Language	Examines the proper use of gender-neutral words, avoidance of sexist terms, and appropriate pronoun selection.

Representation of Women and Men	Evaluates whether both genders are portrayed fairly and proportionally within the study's content, examples, and discussions.
Inclusivity / Stereotyping	Determines whether the paper avoids gender-based generalizations and promotes inclusiveness in its analysis or conclusions.
Authorship and Citations	Reviews the balance of male and female authors in cited works and the recognition of contributions across genders.

Table 1. Key Aspects Considered from the HGDG Checklist

Requirement Documentation

This phase discusses what the software will do. It includes the modules of the proposed system/solution with the description.

Hardware Requirements

The hardware requirements required for the development of the system include an Intel Core i5 processor (10th Generation or higher), 8GB of RAM, and 256GB solid-state drive (SSD). These components are sufficient to support Python-based development, execute natural language processing operations, and train lightweight machine learning models such as Naive Bayes. The processor provides adequate performance for text preprocessing and classification, while the memory capacity allows stable handling of moderate datasets and multitasking during system testing. The SSD ensures faster data access and efficient storage management, contributing to overall smooth and reliable system performance.

Software Requirements

The Automated Assessment of Academic Articles' Gender-Responsiveness will be developed using Python (Flask) for the backend, React.js for the frontend, and Firebase for data storage and user authentication. The development process will be carried out using Visual Studio Code (VS Code) as the main integrated development environment (IDE), chosen for its flexibility and support for both Python and JavaScript-based frameworks.

React.js will serve as the frontend framework responsible for building an interactive and responsive user interface. It will allow users to upload academic papers, receive automated feedback, and view results in real time. React's modular component structure ensures easy maintenance, scalability, and seamless communication with the backend server through API integration.

The backend will be developed using Python with the Flask framework to manage data processing, handle uploaded documents, and connect with machine learning models. The system will employ two key algorithms: Naive Bayes for efficient and lightweight text classification, and BERT (Bidirectional Encoder Representations from Transformers) for deeper contextual analysis of gender-biased language. Together, these models will enable accurate and adaptive assessment of the gender-responsiveness of academic articles.

Firebase will be used as the system's cloud platform, providing secure storage for user data and authentication services. Its integration ensures data consistency, scalability, and reliable backend support without the need for a separate database server, making it ideal for web-based deployment.

System Development

This system will follow the Rapid Application Development software development approach. RAD was chosen for its iterative and component-based nature, emphasizing rapid prototyping and continuous user feedback to refine each part of the system efficiently.

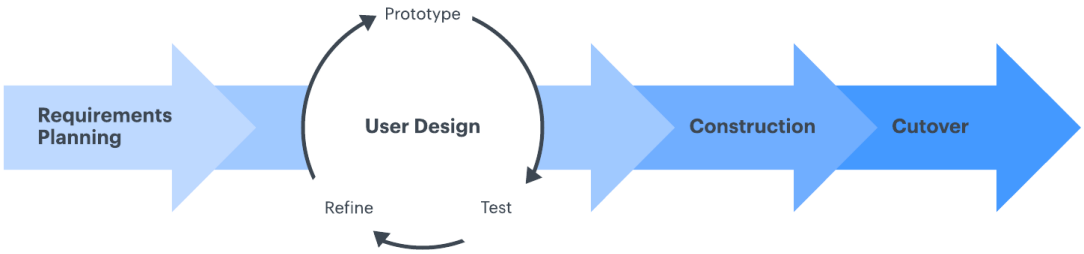


Figure 6. Software methodology of the system using RAD

1. Requirements Planning and User Design

The process begins with Text Extraction, where uploaded files are converted into readable text. The text then undergoes Preprocessing to remove noise and prepare it for analysis. Next, the Classification and Analysis module, where BERT and Naïve Bayes evaluate the text for gender-responsive content. The results are then displayed through a dynamic User Interface that allows users to easily view and interpret findings.

The system is developed using Python (Flask) for backend processing, React.js for the frontend interface, and Firebase for database management and authentication.

2. Dataset Development and Preprocessing

The dataset for this study will be derived from open-access academic papers and research articles, serving as the primary source of textual data. These documents will undergo text extraction, where individual sentences are isolated and used as the main data units for training and testing. Each sentence will be manually labeled based on its degree of gender responsiveness, classified as *gender-responsive*, *non-gender-*

responsive, or *neutral*. This sentence-level labeling approach allows for more accurate detection of gender bias and inclusive language patterns within academic writing. The dataset will then be used to train the Naïve Bayes and BERT models, enabling the system to evaluate uploaded academic papers by analyzing them sentence by sentence.

For the aspects to be followed, relevant terms and keywords were derived from official GAD checklists and gender-sensitive writing guides. These linguistic indicators served as features that allowed the system to classify texts objectively based on their responsiveness level.

Indicators for GAD Features

Each article in the dataset was annotated with indicators reflecting the presence or absence of GAD-related elements. The annotation process involved marking gendered language, inclusive expressions, and authorship patterns. These annotations were cross-checked with reference materials such as the Harmonized Gender and Development Guidelines (HGDG) to ensure accuracy and consistency during model training.

FEATURE CATEGORY	EXAMPLE TERMS / INDICATORS
Gendered Language	He, she, his, her
Inclusive Terms	Everyone, folks, people, researchers
Stereotypical Words	“Women are emotional”, “Men are strong”
Authorship	Balance of female and male in references

Table 4. Sample Linguistic Features

3. Model Building

The Automated Assessment System is designed using a modular structure, where each component performs a specific function contributing to the overall analysis of gender responsiveness in academic papers. The system is divided into several modules, each with their specific contributions to the system.

TEXT EXTRACTION

The text extraction module serves as the entry point of the system, responsible for retrieving and segmenting textual data from uploaded academic papers in formats such as PDF or DOCX. Extracted content is divided into sentences to enable more precise classification, as gender responsiveness is often reflected at the sentence level rather than across an entire document.

PyMuPDF

PyMuPDF reads each page sequentially and converts the document's internal layout structure into readable text. It identifies text blocks and line coordinates, merges them into logical paragraphs, and eliminates layout artifacts such as unnecessary line breaks, multiple spaces, or embedded font data. It also filters out non-text elements like images, annotations, and embedded objects.

python-docx

python-docx accesses the document's XML structure, which stores textual data in a hierarchy of paragraphs and runs. The library iterates through these elements to retrieve only the text portions, excluding non-text elements such as charts, images, or styles. It automatically handles spacing, punctuation normalization, and line continuity between runs, ensuring that text fragments are properly concatenated. The output is clean, linearized text extracted directly from the document body, maintaining the intended formatting and sentence order from the original paper.

Together, these processes ensure that text extraction produces clean and logically ordered content, minimizing errors and inconsistencies before the data enters the system's preprocessing module for further refinement.

PREPROCESSING

The preprocessing module is responsible for refining the extracted text, ensuring it is clean, standardized, and ready for analysis by the machine learning models. Each sentence is processed individually to maintain accuracy in classification.

NLTK (Natural Language Toolkit)

The system uses NLTK to perform the core preprocessing tasks, which transform raw text into a structured format suitable for machine learning models. The processes include:

- **Normalization**: Converts all characters to lowercase to reduce redundancy caused by capitalization (e.g., "He" → "he").

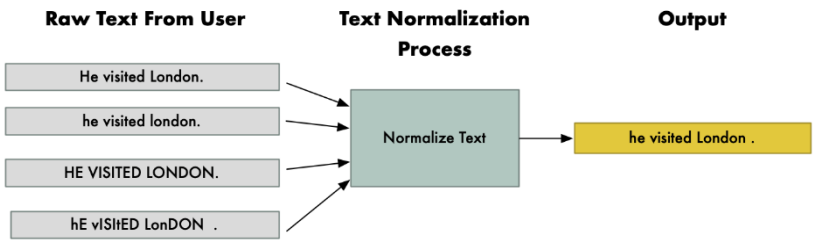


Figure 7.1. Normalization Process

- **Tokenization**: Splits sentences into individual words or tokens, enabling word-level analysis for pattern recognition.

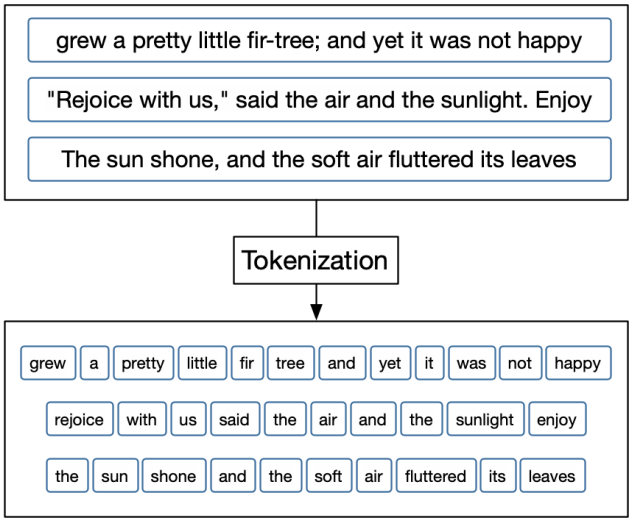


Figure 7.2. Tokenization Process

These tokens then become the features that the model uses to learn patterns. In Naïve Bayes, for instance, each token's frequency or weight contributes to calculating the probability of a sentence belonging to a particular class (e.g., gender-responsive or non-gender-responsive). Tokenization ensures that the text is broken down into consistent and comparable units, making it possible to convert sentences into numerical representations, such as TF-IDF vectors, for training machine learning models.

- **Stop Words Removal**: Eliminates common words such as "the," "is," and "and" that carry little semantic value, reducing noise in the dataset.

Sample text with Stop Words	Without Stop Words
GeeksforGeeks – A Computer Science Portal for Geeks	GeeksforGeeks , Computer Science, Portal ,Geeks
Can listening be exhausting?	Listening, Exhausting
I like reading, so I read	Like, Reading, read

Figure 7.3. Stop Words Removal Example

- **Punctuation Removal**: Removes all punctuation marks to prevent irrelevant characters from affecting model training.

- **Lemmatization**: Converts words to their base forms (e.g., “running” → “run”), ensuring that different word variations are treated consistently.

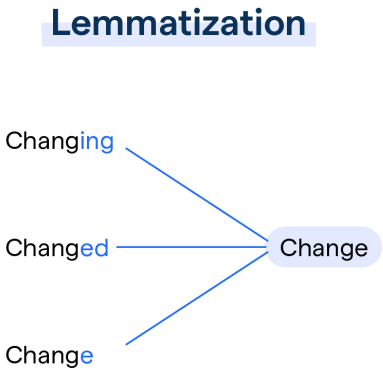


Figure 7.4. Lemmatization Process

After these steps, the cleaned and structured sentences are paired with their corresponding labels, *gender-responsive*, *non-gender-responsive*, or *neutral* and are ready for input into the Naïve Bayes and BERT models.

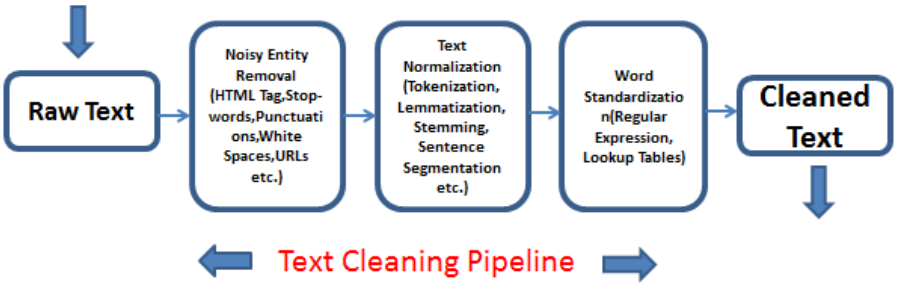


Figure 7.5. Text Preprocessing in Python using spaCy library

FEATURE EXTRACTION

The feature extraction process converts preprocessed text into numerical representations suitable for machine learning models. In this system, ***TF-IDF (Term Frequency–Inverse Document Frequency)*** is used to quantify the importance of each word in a sentence relative to the dataset.

Term Frequency (TF) measures how often a term appears in a sentence compared to the total number of terms in that sentence. It is calculated as:

$$TF(t,s) = \frac{f_{t,s}}{\sum_k f_{k,s}}$$

Inverse Document Frequency (IDF) measures the importance of a term across all sentences by reducing the weight of terms that appear frequently and increasing the weight of rare, informative terms:

$$IDF(t) = log\ \frac{N}{1 + n_t}$$

The TF-IDF value for a term in a sentence is obtained by multiplying the TF and IDF values:

$$TFIDF(t,s) = TF(t,s) \times IDF(t)$$

This results in a numerical vector representation for each sentence, where each dimension corresponds to a term in the vocabulary, and the value reflects its relative importance. These vectors serve as the input features for the Naïve Bayes classifier, enabling it to learn probabilistic relationships between words and sentence categories—*gender-responsive*, *non-gender-responsive*, or *neutral*.

Using TF-IDF ensures that the model emphasizes meaningful and distinctive words, reduces the impact of common terms, and provides a standardized format for machine learning analysis.

N-grams

N-grams are contiguous sequences of *n* items from a given text, where the items are typically words or characters. In natural language processing, n-grams are used to capture patterns, context, and relationships between consecutive words in a sentence.

- **Unigrams (1-gram)**: Single words, representing the basic unit of text.
- **Bi-grams (2-grams)**: Sequences of two consecutive words, which capture simple word pairs and relationships.

- **Tri-grams (3-grams)**: Sequences of three consecutive words, which capture longer phrases and contextual patterns.

By including bi-grams and tri-grams in the feature extraction process, the model can better capture **multi-word expressions and contextual cues**, improving the accuracy of Naïve Bayes and BERT in detecting gender responsiveness.

Example Table of Feature Vectors

Text	Unigrams	Bi-grams	Tri-grams
“Women contribute equally”	[0.5, 0.7, 0.6]	[0.6, 0.5]	[0.7]
“Research must be inclusive”	[0.4, 0.3, 0.5]	[0.5, 0.4]	[0.6]

Table 5. Feature Vectors of Unigrams, Bi-grams, and Tri-grams

This table illustrates how each sentence is converted into a vector representing the importance of unigrams, bi-grams, and tri-grams, providing rich information for classification models.

CLASSIFICATION

The Classification Module is responsible for analyzing the preprocessed and vectorized text to determine the gender responsiveness of each sentence. This module employs two machine learning models—Naïve Bayes and BERT—which complement each other to provide both probabilistic and context-aware classification.

Naïve Bayes

Naïve Bayes is a probabilistic machine learning algorithm based on Bayes’ Theorem, used for classification tasks. It operates under the “naïve” assumption that

all features (words, in the case of text) are independent of each other. Despite this simplification, Naïve Bayes is highly effective for text classification problems because it relies on word frequency and probability to determine the most likely class of a sentence or document.

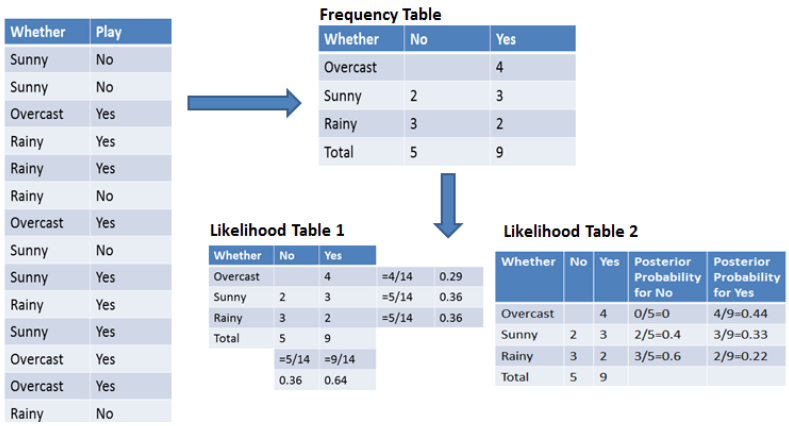
Naïve Bayes performs **probabilistic classification** using the TF-IDF features extracted from each sentence. The processes include:

- **Training**: Learns the likelihood of each word occurring in each class (*gender-responsive*, *non-gender-responsive*, *neutral*).
- **Probability Calculation**: Computes the probability of a sentence belonging to each class using:

$$P(C \mid s) = P(C) \times \prod_{i=1}^{n} P(w_i \mid C)$$

where $P(C)$ is the prior probability of class C and $P(C)$ is the likelihood of word w_i given class C.

- **Prediction**: Selects the class with the highest probability as the sentence label.



Whether	Play
Sunny	No
Sunny	No
Overcast	Yes
Rainy	Yes
Rainy	Yes
Rainy	No
Overcast	Yes
Sunny	No
Sunny	Yes
Rainy	Yes
Sunny	Yes
Overcast	Yes
Overcast	Yes
Rainy	No

Frequency Table

Whether	No	Yes
Overcast		4
Sunny	2	3
Rainy	3	2
Total	5	9

Likelihood Table 1

Whether	No	Yes		
Overcast		4	=4/14	0.29
Sunny	2	3	=5/14	0.36
Rainy	3	2	=5/14	0.36
Total	5	9		
	=5/14	=9/14		
	0.36	0.64		

Likelihood Table 2

Whether	No	Yes	Posterior Probability for No	Posterior Probability for Yes
Overcast		4	0/5=0	4/9=0.44
Sunny	2	3	2/5=0.4	3/9=0.33
Rainy	3	2	3/5=0.6	2/9=0.22
Total	5	9		

Figure 7.5. Naïve Bayes Likelihood Tables Based on Weather Data

In this system, Naïve Bayes performs probabilistic classification using the TF-IDF features extracted from each sentence. TF-IDF transforms each sentence into a vector where each dimension corresponds to a word in the vocabulary, and its value represents the word's relative importance. These vectors are then used to calculate

the probability of the sentence belonging to each class: *gender-responsive*, *non-gender-responsive*, or *neutral*.

Example Likelihood Table:

Word	Gender-Responsive	Non-Gender-Responsive	Neutral
Women	0.3	0.05	0.01
Equality	0.25	0.02	0.03
Research	0.20	0.10	0.05

Table 6.1. Sample Likelihood Table based on Gender Responsiveness

Example Workflow:

Sentence	TF-IDF Vector	Probabilities per Class	Predicted Class
“Women contribute equally”	[0.5, 0.7, 0.6]	Gender-Responsive: 0.063 Non-Gender-Responsive: 0.0007 Neutral: 0.0002	Gender-Responsive

Table 6.2. Sample Workflow based on Gender Responsiveness

Naïve Assumption

The “naïve” assumption treats words as independent, even though in natural language they are often contextually related. This simplification allows efficient computation while still producing accurate results for text classification.

BERT (Bidirectional Encoder Representations from Transformers)

BERT is a deep learning model developed by Google that excels in natural language understanding. Unlike traditional models that read text in a single direction, BERT uses a bidirectional transformer architecture, allowing it to consider both preceding and succeeding words when analyzing a sentence. This enables the model to capture contextual meaning more accurately, including subtle cues and implied semantics.

BERT provides **contextualized classification** by capturing the relationships between words in each sentence. Its processes include:

- **Pretraining and Fine-Tuning**: Utilizes a pre-trained BERT model, fine-tuned on the labeled dataset for gender-responsiveness detection.
- **Tokenization**: Splits sentences into subword tokens and adds special tokens such as [CLS] and [SEP].
- **Embedding and Attention**: Converts tokens into embeddings and applies self-attention to understand bidirectional context.
- **Classification Layer**: Uses the [CLS] token representation in a dense layer with softmax activation to output class probabilities for *gender-responsive*, *non-gender-responsive*, or *neutral*.

By combining Naïve Bayes' statistical efficiency and BERT's contextual understanding, the Classification Module achieves accurate, reliable, and scalable assessment of gender responsiveness in academic sentences.

USER INTERFACE INTERACTIONS

This module facilitates interaction between the user and the system, providing a platform to upload academic papers, view analysis results, and navigate the system. It is developed using **React.js** and is integrated with **Firebase** to manage front-end backend connectivity. The key components of the UI include the file upload component

for submitting PDF or DOCX documents, the results display component for presenting gender responsiveness assessments, and the navigation component for smooth transitions between system features.

System Testing

System testing was conducted to ensure that the developed Automated GAD Responsiveness Assessment System performs accurately and reliably based on the intended objectives. The goal of testing was to verify that all system components including file upload, text extraction, preprocessing, classification, and result generation function as expected under various scenarios.

Functional Testing

Functional Testing evaluates whether the system performs its intended functions according to requirements. Iterative testing ensures that each function works correctly before adding new modules. Tests include:

FUNCTIONS	INPUT	EXPECTED OUTPUT	ACTUAL OUTPUT
Upload PDF or docx file	Sample article in PDF format	Extracted text displayed	
Text Preprocessing	Raw extracted text	Stop words, symbols, and extra spaces removed	
Feature Extraction	Preprocessed text	Feature vectors generated using TF-IDF and n-grams ready for classification	
Classification	Feature Vector	Determines whether the uploaded article is responsive or not	

Report Generation	Processed article	Displays summary and recommendations	

Table 6. Functions, expected, and actual outputs of the system

The primary objective of the system is to automate the assessment of academic articles for Gender and Development (GAD) responsiveness through text extraction, linguistic analysis, and classification using the Naive Bayes algorithm. Specifically:

a) The document upload and extraction component receives academic articles in supported file formats and converts their content into plain, analyzable text.
b) The text preprocessing module refines the extracted text by removing stop words, punctuation, and irrelevant characters, retaining only essential linguistic elements.
c) The feature extraction process detects linguistic features such as gendered terms, pronouns, and inclusivity indicators that contribute to GAD responsiveness assessment.
d) The Naive Bayes classification module evaluates the extracted features and determines whether each academic article is categorized as “Responsive” or “Non-Responsive.”
e) The report generation and visualization module produces a clear, structured summary of the classification results along with relevant recommendations.

Model Testing

Model Testing evaluates the performance and reliability of the trained classification models used in the system. It ensures that both the Naive Bayes and BERT models accurately identify GAD-responsive and non-responsive content based on predefined key aspects.

Confusion Matrix

The confusion matrix is a matrix of numbers that tell us where a model gets confused. It is a class-wise distribution of the predictive performance of a classification model, that is, the confusion matrix is an organized way of mapping the predictions to the original classes to which the data belong (Kundu R. 2022).

The confusion matrix is a performance evaluation tool that summarizes the results of a classification model. It shows how many predictions were correct or incorrect by comparing the model's outputs with the actual labels.

Elements of the Confusion Matrix:

- **True Positive (TP):** Instances where the model correctly predicts a positive class.
- **True Negative (TN):** Instances where the model correctly predicts a negative class.
- **False Positive (FP):** Instances where the model incorrectly predicts a positive class when it is actually negative.
- **False Negative (FN):** Instances where the model incorrectly predicts a negative class when it is actually positive.

From this matrix, **performance metrics** are calculated:

- **Accuracy:** Overall correctness of predictions.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

- **Precision:** Correctness of positive predictions.

$$Precision = \frac{TP}{TP + FP}$$

- **Recall:** Coverage of actual positive instances.

$$Recall = \frac{TP}{TP + FN}$$

- **F1-Score:** Harmonic mean of Precision and Recall.

$$F1\ Score = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

This evaluation shows which sentences are correctly classified and identifies areas for improvement, such as misclassified Neutral sentences. BERT is also validated using the same metrics, often showing higher precision and recall due to its contextual understanding.

System Evaluation

The system was evaluated based on modified ISO/IEC 25010 Software and Data Quality Evaluation Instrument. This is in terms of functional suitability, performance efficiency, compatibility, interaction capability, reliability, security, maintainability, flexibility, and safety as shown in Table 7.

Table 7. Software and data quality evaluation instrument

INDICATOR	5	4	3	2	1
A. Functional Suitability					
1. **Functional completeness** - degree to which the set of functions covers all the specified tasks and intended users' objectives.					
2. **Functional correctness** - degree to which a product or system provides accurate results when used by intended users.					
3. **Functional appropriateness** - degree to which the functions facilitate the accomplishment of specified tasks and objectives.					
B. Performance Efficiency					
1. **Time behaviour** - degree to which the response time and throughout rates of a product or system, when performing its functions, meet requirements.					
2. **Resource utilization** - degree to which the amounts and types of resources used by a product or system, when performing its functions, meet requirements.					
3. **Capacity** - degree to which the maximum limits of a product or system parameter meet requirements.					
C. Compatibility					
1. **Interoperability** - degree to which a system, product or component can exchange information with other products and mutually use the information that has been exchanged.					
D. Interaction Capability					
1. **Appropriateness recognizability** - degree to which users can recognize whether a product or system is appropriate for their needs.					

2. **Learnability** - degree to which the functions of a product or system can be learnt to be used by specified users within a specified amount of time.
3. **Operability** - degree to which a product or system has attributes that make it easy to operate and control.
4. **User error protection** - degree to which a system prevents users against operation errors.
5. **User engagement** - degree to which a user interface presents functions and information in an inviting and motivating manner encouraging continued interaction.
6. **Inclusivity** - degree to which a product or system can be used by people of various backgrounds (such as people of various ages, abilities, cultures, ethnicities, languages, genders, economic situations, etc.).
7. **Self-descriptiveness** - degree to which a product presents appropriate information, where needed by the user, to make its capabilities and use immediately obvious to the user without excessive interactions with a product or other resources (such as user documentation, help desks or other users).

E. Reliability

1. **Faultlessness** - degree to which a system, product or component performs specified functions without fault under normal operation.
2. **Availability** - degree to which a system, product or component is operational and accessible when required for use.
3. **Recoverability** - degree to which, in the event of an interruption or a failure, a product or system can recover the data directly affected and re-establish the desired state of the system.

F. Security

1. **Confidentiality** - degree to which a product or system ensures that data are accessible only to those authorized to have access.
2. **Integrity** - degree to which a system, product or component ensures that the state of its system and data are protected from unauthorized modification or deletion either by malicious action or computer error.
3. **Authenticity** - degree to which the identity of a subject or resource can be proved to be the one claimed.

G. Maintainability

1. **Modularity** - degree to which a system or computer program is composed of discrete components such that a change to one component has minimal impact on other components.
2. **Modifiability** - degree to which a product or system can be effectively and efficiently modified without introducing defects or degrading existing product quality.

3. **Testability** - degree of effectiveness and efficiency with which test criteria can be established for a system, product or component and tests can be performed to determine whether those criteria have been met.

H. Flexibility

1. **Adaptability** - degree to which a product or system can effectively and efficiently be adapted for or transferred to different hardware, software or other operational or usage environments.
2. **Scalability** - degree to which a product can handle growing or shrinking workloads or to adapt its capacity to handle variability.

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